

AE4 control of salivary secretion nearly a decade later: acid–base balance, transporter stoichiometry and missing network regulation

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Abstract

Nearly a decade ago, we constructed a mathematical model to explain why deletion of the cation-dependent chloride–bicarbonate exchanger AE4 impairs salivary secretion, whereas deletion of AE2 has little effect. That model reproduced a substantial AE4-null secretory deficit and suggested that AE4’s monovalent-cation coupling was the essential distinction. Here we return to that mechanism using a conservation-explicit model reconstructed from the intervening experimental and modelling literature. The new model evolves intracellular and luminal sodium, potassium, chloride, total inorganic carbon, total alkalinity and volume. It includes finite carbonate buffering, source-fixed NKCC1 and NHE1 kinetics, electrogenic sodium–bicarbonate cotransport, beta-adrenergic regulation of AE4 and explicit current, charge, carbon and water closure.

The reconstruction changes the interpretation of the original result. An inward sodium–bicarbonate pathway is required for AE4 to carry a substantial stimulated chloride load without producing an acid–base contradiction. In the reconstructed wild type, NKCC1 and AE4 supply 79.9% and 20.1% of positive chloride loading, respectively, while pH remains physiological. However, replacing the original concentration-weighted AE4 cation routing by equal sodium and potassium routing produces only a 3.86% secretion deficit after complete AE4 loss because NKCC1 compensation replaces most of the missing chloride input. We then test seven source classes permitted by the molecular experiments of Catalán et al. (2025), with all predictions frozen before comparison with the held-out phenotype. None recovers the approximately 35% experimental deficit. Several carbonate classes instead fail the wild-type pH gate during stimulation.

Finally, an inverse construction shows that an additional AE4-dependent reduction in stimulated CaCC recruitment can produce a 30.3% null deficit while retaining the declared 600-s physiological gates. This coupling was selected against the target and is not independent validation. The combined results show that the biological importance of AE4 survives the decade, but transporter stoichiometry alone does not explain it. Acid–base supply and sustained network regulation are the missing parts of the original mechanism.

Keywords: salivary secretion; AE4; SLC4A9; acid–base balance; sodium–bicarbonate co-transport; NKCC1; mathematical physiology

1 Introduction

Saliva is produced by the acinar cells of the major salivary glands. Its functions extend from lubrication and digestion to the protection of the oral cavity, and a sustained reduction in flow has substantial clinical consequences. The central physical mechanism has been understood for many years: chloride is accumulated above electrochemical equilibrium in the acinar cell, released across the apical membrane after stimulation, and accompanied by cation and water

movement into the lumen [Silva et al., 1977, Gin et al., 2007, Palk et al., 2010, Melvin et al., 2005]. The biological difficulty lies not in stating this picture, but in accounting quantitatively for the many transport processes that maintain it.

At the gland level, this is the first stage of a two-stage secretory process. Acinar cells generate the primary electrolyte and fluid secretion, whereas the duct system subsequently modifies its ionic composition before saliva reaches the oral cavity [Thaysen et al., 1954, Young and Schögel, 1966, Melvin et al., 2005]. The model studied here addresses the acinar stage. Organ-level submandibular flow is used as the phenotype to be explained, but ductal ion transport and modification of the final saliva are not represented explicitly.

NKCC1 is the dominant basolateral chloride-loading pathway in salivary acinar cells: pharmacological inhibition and genetic loss markedly depress secretion, yet leave a substantial residual response, motivating the long-standing proposal of a parallel bicarbonate-dependent chloride uptake route [Martinez and Cassity, 1983, Evans et al., 2000, Nauntofte, 1992, Nguyen et al., 2004]. In 2015, Peña-Münzenmayer et al. reported an initially surprising result. Deletion of the cation-dependent chloride–bicarbonate exchanger AE4 reduced stimulated salivary secretion by approximately 35%, whereas deletion of the more widely expressed exchanger AE2 had little effect [Peña Münzenmayer et al., 2015]. The experiment also showed a lower resting intracellular chloride concentration and slower chloride reuptake in AE4-deficient acinar cells. The early secretion response was comparatively well preserved, while the difference between wild type and knockout developed during sustained stimulation. These findings raised a straightforward question: why should AE4 be so important when NKCC1 remains the dominant chloride-loading pathway and AE2 can, at least superficially, perform the same chloride–bicarbonate exchange?

Our 2018 model was built to answer that question [Vera-Sigüenza et al., 2018]. It extended earlier single-cell secretion models by adding AE2, NHE1 and a cation-dependent AE4 cycle. The model reproduced a negligible AE2-null effect and an approximately 24% reduction in secretion after AE4 deletion. It attributed the distinction to AE4’s ability to couple chloride and bicarbonate exchange to monovalent-cation transport. In the model, loss of AE4 increased intracellular sodium and potassium, limited NHE1 and pump activity, reduced chloride accumulation and lowered secretion. It also predicted only minimal activation of NKCC1 after either AE knockout, consistent with the isolated NKCC1 measurements available at the time [Peña Münzenmayer et al., 2015, Vera-Sigüenza et al., 2018]. The result gave mathematical support to the experimental conclusion that AE4, rather than AE2, supplies the physiologically important bicarbonate-dependent chloride-loading pathway.

That explanation was useful, but it was also incomplete. The 2018 model used a deliberately compact acid–base system. Intracellular bicarbonate was supplied only through carbon dioxide hydration, while NHE1 extruded protons and AE2 and AE4 exported bicarbonate. No additional proton or bicarbonate transporter was included. AE4 was represented as a $1\text{Cl}^- : 1\text{ monovalent cation} : 2\text{HCO}_3^-$ cycle, with the sodium and potassium components assigned according to their intracellular concentrations. The model also reached its reduced knockout flow within about two minutes, considerably faster than the roughly ten-minute experimental development of the phenotype. Most importantly, the original agreement with the organ-level flow deficit did not establish that the proposed intracellular mechanism was unique.

A decade of experiments has changed what can reasonably be asked of the model. AE4 has been shown to be an electroneutral cation-dependent chloride–bicarbonate exchanger, with both sodium and potassium supporting transport [Peña Münzenmayer et al., 2016]. Beta-adrenergic stimulation activates AE4 through the cAMP/PKA pathway, with the S173 region implicated in the response [Peña Münzenmayer et al., 2021]. More recently, Catalán et al. identified molecular determinants that distinguish sodium-supported from potassium-

supported transport and proposed several possible whole-cycle stoichiometries [Catalán et al., 2025]. These experiments constrain AE4 much more strongly than was possible in 2018, but they do not determine a complete kinetic law for the secreting acinar cell.

There have also been important developments in the surrounding secretion model. Apical potassium channels and sodium–potassium pumps alter the cation topology of secretion [Almássy et al., 2018]. Spatial and multicellular work has shown when the well-stirred cell approximation is adequate and when calcium and volume feedback cannot be separated [Vera-Sigüenza et al., 2019, Pages et al., 2019, Vera-Sigüenza et al., 2020, Sneyd et al., 2021]. These results make it possible to return to the AE4 problem with a more explicit accounting of conserved quantities, current closure and transporter provenance.

The aim of the present study is therefore not to reproduce the 2018 answer with newer code. We ask instead which parts of the original mechanism survive reconstruction. First, we rebuild the model in conserved sodium, potassium, chloride, total inorganic carbon and total alkalinity coordinates. This exposes a previously hidden acid–base restriction: substantial productive AE4 chloride loading consumes alkalinity faster than the original NHE1–CO₂ cycle can replace it. We show that a stimulus-recruited electrogenic sodium–bicarbonate pathway supplies the missing degree of freedom and permits AE4 to carry a substantial fraction of wild-type chloride loading while maintaining pH. Second, we test whether the original cation-routing argument survives the reconstructed network. Third, we use the mechanism classes proposed by Catalán et al. as a finite, predeclared panel and freeze their predictions before comparison with the held-out secretion phenotype. Finally, we use an explicitly target-selected construction to identify the strength and type of additional coupling that would be required if stoichiometry alone is insufficient.

The story that emerges is both more modest and more informative than the original one. AE4 remains a distinct and important component of the secretory network. However, its cation stoichiometry does not by itself determine the knockout phenotype. The decisive constraints lie in acid–base supply, NKCC1 compensation and the regulation of sustained apical chloride exit.

2 Model

The reconstructed model describes one acinar cell, a finite luminal compartment and an infinite extracellular bath. It retains the well-stirred approximation used in the original AE4 model, but replaces several concentration equations by conserved amount balances. This change is essential for acid–base chemistry, because internal protonation reactions may redistribute carbon species without creating carbon or alkalinity. A schematic comparison is shown in figure 1.

2.1 Compartments, amounts and concentrations

Let $n_{s,i}$ and $n_{s,l}$ denote the intracellular and luminal amounts of ion $s \in \{\text{Na}^+, \text{K}^+, \text{Cl}^-\}$. The cell and lumen volumes are V_i and V_l , and concentrations are obtained from

$$[s]_r = \frac{n_{s,r}}{V_r}, \quad r \in \{i, l\}. \quad (1)$$

The numerical units are fmol for amount, pL for volume and mM for concentration, so the conversion in equation (1) is exact. The bath concentrations remain constant.

Rather than evolve free bicarbonate and proton concentrations independently, we evolve the total inorganic carbon amount $n_{T,r}$ and total alkalinity amount $n_{A,r}$ in each finite

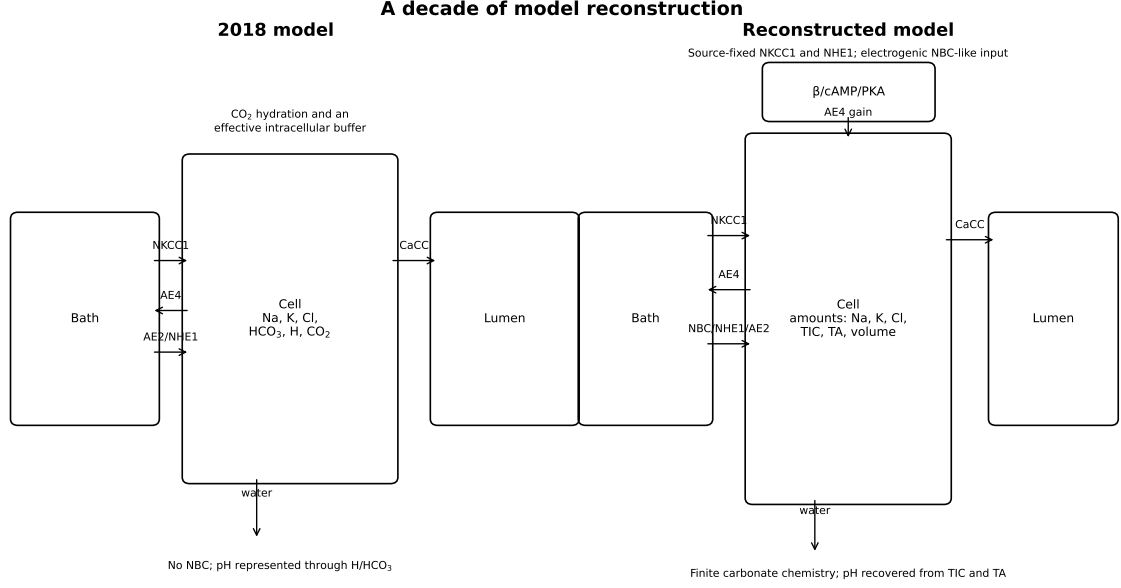


Figure 1: Evolution from the 2018 model to the conservation-explicit reconstruction. The original model tracked explicit bicarbonate, protons and carbon dioxide, but included no sodium–bicarbonate cotransporter. The reconstructed model evolves total inorganic carbon (TIC) and total alkalinity (TA), recovers pH algebraically, includes an electrogenic NBC-like input and uses source-fixed NKCC1 and NHE1 laws together with beta/cAMP/PKA regulation of AE4.

compartment. The complete core state is therefore

$$y = (n_{Na,i}, n_{K,i}, n_{Cl,i}, n_{T,i}, n_{A,i}, V_i, n_{Na,l}, n_{K,l}, n_{Cl,l}, n_{T,l}, n_{A,l}, V_l), \quad (2)$$

augmented by the regulatory states described below. This amount formulation makes dilution, charge accounting and water transport explicit.

2.2 Carbonate chemistry and intracellular pH

Total inorganic carbon is the sum of dissolved carbon dioxide, bicarbonate and carbonate,

$$T = [CO_2] + [HCO_3^-] + [CO_3^{2-}].$$

We include one finite monoprotic non-carbonate buffer with total concentration $B_T = [BH] + [B^-]$. Total alkalinity is

$$A = [HCO_3^-] + 2[CO_3^{2-}] + [B^-] + [OH^-] - [H^+]. \quad (3)$$

For fixed T , A , B_T and volume, pH is recovered by solving equation (3) together with the two carbonate dissociation equilibria, water ionisation and the buffer equilibrium. If K_1 , K_2 and $H = [H^+]$ denote the carbonate equilibrium constants and proton concentration, the carbon fractions are

$$\begin{aligned} \alpha_{CO_2} &= \frac{H^2}{H^2 + K_1 H + K_1 K_2}, \\ \alpha_{HCO_3^-} &= \frac{K_1 H}{H^2 + K_1 H + K_1 K_2}, \\ \alpha_{CO_3^{2-}} &= \frac{K_1 K_2}{H^2 + K_1 H + K_1 K_2}. \end{aligned}$$

Thus $[\text{CO}_2] = T\alpha_{\text{CO}_2}$, $[\text{HCO}_3^-] = T\alpha_{\text{HCO}_3^-}$ and $[\text{CO}_3^{2-}] = T\alpha_{\text{CO}_3^{2-}}$. Fast internal hydration and protonation reactions conserve both T and A . External species fluxes are mapped to these coordinates. One bicarbonate adds one unit of carbon and one equivalent of alkalinity, one carbonate adds one unit of carbon and two equivalents of alkalinity, and proton extrusion raises alkalinity by one equivalent.

This formulation is the first material departure from the 2018 acid–base model. There, carbon dioxide hydration produced free bicarbonate and protons, NHE1 removed protons, and AE2 and AE4 exported bicarbonate. No additional bicarbonate uptake pathway was present [Vera-Sigüenza et al., 2018]. In the present model, the same biological processes are retained but their effects are expressed in conserved coordinates, which reveals directly whether a proposed transporter can support sustained AE4 activity without violating alkalinity balance.

2.3 Whole-cell amount balances

We use positive fluxes in the directions specified below. J_N is the inward NKCC1 cycle flux, J_H is the inward-sodium/outward-proton NHE1 flux, J_2 is the inward-chloride/outward-bicarbonate AE2 flux, J_B is the inward electrogenic NBC cycle flux, and S_s^4 is the AE4 source for species or conserved quantity s . P_a and P_b are apical and basolateral sodium–potassium pump cycle rates. $J_{K,a}$ and $J_{K,b}$ are potassium effluxes from cell to lumen and bath, and $J_{\text{Cl},a}$ is apical chloride efflux. With these conventions, the intracellular balances are

$$\begin{aligned}\dot{n}_{\text{Na},i} &= J_N + J_H + J_B + S_{\text{Na}}^4 - 3(P_a + P_b), \\ \dot{n}_{\text{K},i} &= J_N + S_{\text{K}}^4 + 2(P_a + P_b) - J_{K,a} - J_{K,b}, \\ \dot{n}_{\text{Cl},i} &= 2J_N + J_2 + S_{\text{Cl}}^4 - J_{\text{Cl},a}, \\ \dot{n}_{T,i} &= J_{\text{CO}_2,b} + J_{\text{CO}_2,a} - J_2 + S_T^4 + 2J_B, \\ \dot{n}_{A,i} &= J_H - J_2 + S_A^4 + 2J_B, \\ \dot{V}_i &= q_b - q_a.\end{aligned}\tag{4}$$

Here $J_{\text{CO}_2,b}$ is carbon dioxide transport from bath to cell and $J_{\text{CO}_2,a}$ is transport from lumen to cell. The corresponding luminal balances are

$$\begin{aligned}\dot{n}_{\text{Na},l} &= 3P_a - J_{\text{Na}}^p - q_{\text{out}}[\text{Na}^+]_l, \\ \dot{n}_{\text{K},l} &= -2P_a + J_{K,a} - J_{\text{K}}^p - q_{\text{out}}[\text{K}^+]_l, \\ \dot{n}_{\text{Cl},l} &= J_{\text{Cl},a} - J_{\text{Cl}}^p - q_{\text{out}}[\text{Cl}^-]_l, \\ \dot{n}_{T,l} &= -J_{\text{HCO}_3^-}^p - J_{\text{CO}_2,a} - q_{\text{out}}T_l, \\ \dot{n}_{A,l} &= -J_{\text{HCO}_3^-}^p - q_{\text{out}}A_l, \\ \dot{V}_l &= q_a + q_t - q_{\text{out}}.\end{aligned}\tag{5}$$

The paracellular fluxes J_s^p are positive from lumen to bath. The lumen is finite, and therefore secretion removes both water and dissolved solutes through the outflow terms in equation (5).

2.4 Basolateral chloride loading and pH regulation

2.4.1 NKCC1

NKCC1 transports one sodium, one potassium and two chloride ions inward per cycle. We use the two-state Palk–Benjamin steady flux [Benjamin and Johnson, 1997, Palk et al., 2010], corrected to the concentration units used here,

$$J_N = \alpha_N m_N(t) \frac{a_1 - a_2[\text{Na}^+]_i[\text{K}^+]_i[\text{Cl}^-]_i^2}{a_3 + a_4[\text{Na}^+]_i[\text{K}^+]_i[\text{Cl}^-]_i^2}.\tag{6}$$

The factor α_N is fixed from the accepted wild-type resting state. The activity multiplier $m_N(t)$ is one at rest and rises algebraically with the normalised calcium stimulus to 1.75 at full activation. This modifies capacity, not stoichiometry or reversal.

2.4.2 NHE1

NHE1 exchanges one extracellular sodium for one intracellular proton. We use the eight-state reduction of Cha et al. [2009]. Let

$$\begin{aligned} a &= k_1^+ \frac{[\text{Na}^+]_e}{[\text{Na}^+]_e + K_{Na,o}} \frac{K_{H,o}}{[\text{H}^+]_e + K_{H,o}}, \\ b &= k_2^+ \frac{[\text{H}^+]_i}{[\text{H}^+]_i + K_{H,i}} \frac{K_{Na,i}}{[\text{Na}^+]_i + K_{Na,i}}, \\ c &= k_1^- \frac{[\text{Na}^+]_i}{[\text{Na}^+]_i + K_{Na,i}} \frac{K_{H,i}}{[\text{H}^+]_i + K_{H,i}}, \\ d &= k_2^- \frac{[\text{H}^+]_e}{[\text{H}^+]_e + K_{H,o}} \frac{K_{Na,o}}{[\text{Na}^+]_e + K_{Na,o}}. \end{aligned}$$

The exchanger flux is

$$J_H = N_H m_H(t) M_2 \frac{ab - cd}{a + b + c + d}, \quad M_2 = \left[1 + \left(1 + \frac{[\text{H}^+]_e}{K_{M,o}} \right) \left(\frac{K_{M,i}}{[\text{H}^+]_i} \right)^3 \right]^{-1}. \quad (7)$$

N_H is the wild-type carrier amount obtained from the resting pH reconstruction, while $m_H(t)$ rises from one at rest to 2.3 under full stimulation. One positive cycle adds one sodium amount and one alkalinity equivalent to the cell.

2.4.3 AE2

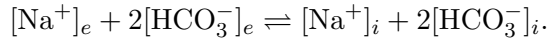
AE2 is represented by a reversible chloride–bicarbonate exchange law,

$$J_2 = G_2 \tanh\left(\frac{A_2}{w}\right), \quad A_2 = \log\left(\frac{[\text{Cl}^-]_e [\text{HCO}_3^-]_i}{[\text{Cl}^-]_i [\text{HCO}_3^-]_e}\right),$$

where positive flux imports chloride and exports bicarbonate. It contributes $+J_2$ to intracellular chloride and $-J_2$ to both TIC and TA.

2.4.4 Stimulus-recruited sodium–bicarbonate cotransport

The reconstructed model includes a minimal electrogenic NBC-like pathway,



This is a structural pathway rather than an assignment to a particular salivary SLC4 isoform. Its inward affinity and flux are

$$\begin{aligned} A_B &= \log\left(\frac{[\text{Na}^+]_e [\text{HCO}_3^-]_e^2}{[\text{Na}^+]_i [\text{HCO}_3^-]_i^2}\right) + \frac{V_b}{V_T}, \\ J_B &= u(t) G_B \tanh\left(\frac{A_B}{w_B}\right). \end{aligned} \quad (8)$$

Here $V_T = RT/F$, and $V_b = \phi_i - \phi_e$ is the basolateral potential. The activation $u(t)$ is exactly zero at the resting calcium concentration and one at the standard stimulated calcium input.

Thus the accepted resting model is an exact nesting limit. Each inward cycle adds one sodium amount, two units of TIC and two alkalinity equivalents. It carries one net negative charge into the cell, so the conventional current is $I_B = FJ_B$ in the cell-to-bath current convention and is included in basolateral current closure.

The need for equation (8) follows directly from equation (4). Without NBC, the sustained alkalinity balance is

$$0 = J_H - J_2 + S_A^4.$$

For the original AE4 stoichiometry, $S_A^4 = -2J_4$. Substantial AE4 chloride loading therefore requires either a comparably large NHE1 flux or another inward source of alkalinity. The NHE1 flux compatible with the measured pH and sodium state is too small to supply that requirement. With NBC, the sustained balance becomes

$$0 = J_H + 2J_B - J_2 + S_A^4, \quad (9)$$

which supplies precisely the missing degree of freedom without importing chloride directly.

2.5 AE4 transport and regulation

The parent AE4 flux is a reversible two-conformation, shared-carrier quasi-steady model. It has one common chloride transition and separate sodium- and potassium-loaded return transitions. For cation $c \in \{\text{Na}^+, \text{K}^+\}$, the forward affinity of the original $1\text{Cl}^- : 1c : 2\text{HCO}_3^-$ cycle is

$$A_{4,c} = \log\left(\frac{[\text{Cl}^-]_e}{[\text{Cl}^-]_i}\right) + \log\left(\frac{[c]_i}{[c]_e}\right) + 2\log\left(\frac{[\text{HCO}_3^-]_i}{[\text{HCO}_3^-]_e}\right). \quad (10)$$

The effective forward and reverse rates are constructed symmetrically, $k^+ = \bar{k} \exp(A/2)$ and $k^- = \bar{k} \exp(-A/2)$, so local detailed balance and the exact reversal condition are preserved. If p_o and p_i are the two carrier occupancies, the cation branch fluxes are

$$\begin{aligned} J_{4,\text{Na}^+} &= N_4 m_4(t) (k_{\text{Na}^+}^+ p_i - k_{\text{Na}^+}^- p_o), \\ J_{4,\text{K}^+} &= N_4 m_4(t) (k_{\text{K}^+}^+ p_i - k_{\text{K}^+}^- p_o), \end{aligned}$$

and the total chloride-loading cycle is $J_4 = J_{4,\text{Na}^+} + J_{4,\text{K}^+}$.

Beta-adrenergic regulation is represented by one effective activation fraction x_4 ,

$$\dot{x}_4 = \frac{\beta(t) - x_4}{\tau_4}, \quad m_4(t) = 1 + g_4 x_4, \quad (11)$$

with $\tau_4 = 30$ s and $g_4 = 0.25$. This is the smallest dynamic representation consistent with the experimentally established beta/cAMP/PKA direction of regulation [Peña Münzenmayer et al., 2021]; it does not assert that the intermediate kinetics have been measured.

For the equal-routing model, the total scalar cycle J_4 is retained but its cation source is divided equally between sodium and potassium. The conserved-coordinate source is

$$(S_{\text{Na}}^4, S_{\text{K}}^4, S_{\text{Cl}}^4, S_T^4, S_A^4) = \left(-\frac{1}{2}, -\frac{1}{2}, 1, -2, -2\right) J_4. \quad (12)$$

This modification isolates cation routing from total AE4 chloride loading. It is not claimed to be a microscopic AE4 mechanism.

Table 1: Predeclared AE4 source classes. Each row gives $(S_{Na}^4, S_K^4, S_{Cl}^4, S_T^4, S_A^4)/J_4$.

Class	Proposed cycle	Source vector	Affinity
C0	equal-routing 1 : 1 : 2 control	$(-1/2, -1/2, 1, -2, -2)$	opposed
C1	$Cl^- + Na^+$ in, $K^+ + HCO_3^-$ out	$(1, -1, 1, -1, -1)$	supported
C2	$Cl^- + Na^+$ in, $2K^+ + CO_3^{2-}$ out	$(1, -2, 1, -1, -2)$	supported
C3a	Cl^- in, $K^+ + 2HCO_3^-$ out	$(0, -1, 1, -2, -2)$	supported
C3b	Cl^- in, $Na^+ + 2HCO_3^-$ out	$(-1, 0, 1, -2, -2)$	opposed
C4a	Cl^- in, $K^+ + CO_3^{2-}$ out	$(0, -1, 1, -1, -2)$	supported
C4b	Cl^- in, $Na^+ + CO_3^{2-}$ out	$(-1, 0, 1, -1, -2)$	opposed

2.6 Catalán 2025 mechanism classes

Catalán et al. experimentally constrained the residues and cation dependence of human AE4, and proposed several possible electroneutral transport cycles [Catalán et al., 2025]. The exact stoichiometries remain hypotheses. We therefore test a finite panel by keeping the inherited scalar $J_4(y, t)$ fixed and changing only the transported source vector. This first-stage construction asks what the proposed stoichiometries do to the whole-cell balances; it does not replace the missing class-specific kinetic law.

The carbonate classes distinguish carbon from alkalinity: outward transport of one CO_3^{2-} removes one unit of TIC and two equivalents of TA. Representing carbonate as bicarbonate would therefore violate both the chemistry and charge balance. All seven source vectors in table 1 are exactly electroneutral and reverse consistently when J_4 changes sign.

2.7 Membrane currents and voltage closure

The apical and basolateral membrane potentials are $V_a = \phi_i - \phi_l$ and $V_b = \phi_i - \phi_e$. The transepithelial potential is $V_t = V_b - V_a$. For ion s of valence z_s , the Nernst potential between the named compartments is

$$E_s = \frac{V_T}{z_s} \log \left(\frac{[s]_{out}}{[s]_{in}} \right).$$

Calcium-activated potassium and chloride conductances use a Hill gate

$$h_{Ca} = \frac{[Ca^{2+}]_i^r}{[Ca^{2+}]_i^r + K_{Ca}^r},$$

and Ohmic currents $I = Gh_{Ca}(V - E)$. Total pump capacity is partitioned between apical and basolateral membranes. For either membrane side r ,

$$P_r = P_{r,max} \frac{[Na^+]_i^3}{[Na^+]_i^3 + K_{Na}^3} \frac{[K^+]_r^2}{[K^+]_r^2 + K_K^2}.$$

Each cycle exports three intracellular sodium ions and imports two potassium ions from the corresponding external compartment.

Paracellular currents are linear,

$$I_s^p = G_s^p(V_t - E_s^p),$$

with positive direction from lumen to bath. Membrane capacitance is fast relative to secretion, so voltages satisfy quasi-steady current closure,

$$\begin{aligned} I_a - I_p &= 0, \\ I_b + I_B + I_p &= 0, \end{aligned} \tag{13}$$

where I_B is omitted at rest because the NBC increment is then exactly zero. The voltage system is solved at every right-hand-side evaluation. Source-charge identities are checked independently against equation (13).

2.8 Water transport and luminal outflow

Water follows osmotic differences between the three compartments,

$$\begin{aligned} q_b &= L_b(\Pi_i - \Pi_e), & q_a &= L_a(\Pi_l - \Pi_i), \\ q_t &= L_t(\Pi_l - \Pi_e), & q_{\text{out}} &= k_{\text{out}} \max(V_l - V_{\text{dead}}, 0). \end{aligned}$$

The osmolarities include sodium, potassium, chloride, total inorganic carbon and the fixed intracellular osmolyte pool. The resulting volume equations are already included in equations (4) and (5). The outflow law is a finite-lumen compliance closure. It is used consistently across all comparisons and is not interpreted as an independently measured ductal pressure law.

2.9 Stimulus, genotypes and numerical protocol

The standard production stimulus combines muscarinic and beta-adrenergic activation for 600 s. Intracellular calcium is prescribed by the experimental protocol, as in the original single-cell model, while the beta input drives equation (11). Wild type has AE4 expression $e_4 = 1$, the reduced-expression case has $e_4 = 0.05$, and the null has $e_4 = 0$. Unless explicitly stated, all genotypes begin from the same frozen wild-type resting state and all non-AE4 parameters remain fixed.

The system is integrated with an implicit Radau method. The numerical gates require positive amounts and volumes, intracellular sodium, potassium, chloride and pH within the predeclared physiological ranges, exact source-charge bookkeeping, current closure, carbon accounting and water conservation. The Catalán mechanism panel was fixed before calculation. All seven class predictions were committed at checkpoint 4cb75308... before comparison with the held-out AE4 secretion phenotype. No class was tuned, discarded or rescued on the basis of its secretion prediction.

3 Results

3.1 Reconstructing wild-type secretion exposes an alkalinity constraint

The 2018 model reproduced the central experimental distinction between AE4 and AE2 deletion, but its wild-type AE4 flux was embedded in an acid-base architecture with no inward bicarbonate transporter. When the model is written in TIC and TA coordinates, the restriction becomes immediate. For the original $1\text{Cl}^- : 1 \text{ cation} : 2\text{HCO}_3^-$ AE4 cycle, the alkalinity equation in equation (4) contains the term $-2J_4$. NHE1 supplies only J_H alkalinity equivalents per unit time. Consequently, increasing productive AE4 chloride loading without an additional source of base forces one of three outcomes: an unphysiological pH shift, an unphysiological NHE1 flux, or a small AE4 contribution.

The stimulus-recruited NBC-like pathway in equation (8) resolves this structural conflict. It supplies two bicarbonate equivalents per inward cycle and is exactly absent at rest. The accepted resting state is therefore unchanged, while stimulated AE4 loading can increase without violating the resting pH calibration. The resulting wild-type trajectory passes the full physiology and conservation gates. Intracellular pH remains between 6.9100 and 7.0348 during the 600-s stimulus, sodium remains below 21.83 mM, chloride remains between 54.08 and 60.30 mM, and cell volume remains below 1.48 pL.

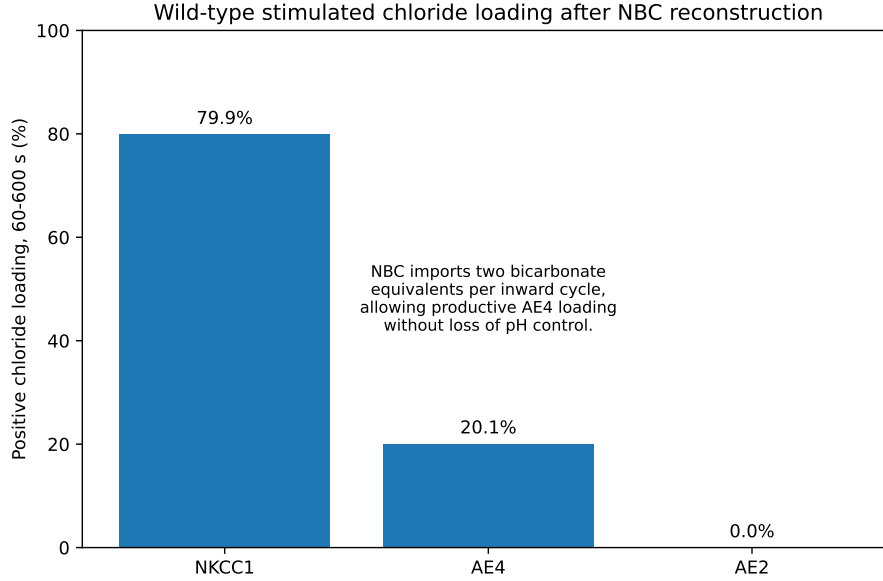


Figure 2: Positive wild-type chloride loading integrated over 60–600 s after the acid–base reconstruction. NKCC1 remains the dominant pathway, while AE4 contributes approximately 20% and AE2 contributes no positive loading. NBC supports this partition by replenishing the alkalinity consumed by AE4.

Over the sustained 60–600 s interval, positive chloride loading is divided between NKCC1 and AE4 in the proportions 79.9248% and 20.0752%, while AE2 contributes no positive loading. The partition is shown in figure 2. NKCC1 remains dominant, as required by the experimental literature, but AE4 is no longer a negligible residual pathway. The essential role of NBC is indirect: it does not import chloride, but permits AE4 to consume intracellular bicarbonate while the cell maintains its alkalinity and pH.

3.2 Equal sodium and potassium routing does not preserve the 2018 knockout mechanism

The original model routed the AE4-coupled monovalent cation according to intracellular sodium and potassium abundance. To isolate whether this routing assumption generated the knockout phenotype, we retained the same total scalar AE4 cycle and imposed the fixed source in equation (12). The modification changes neither total AE4 chloride loading nor bicarbonate export.

All three trajectories, wild type, 5% AE4 and complete AE4 loss, completed 600 s without numerical retry and passed the declared physiology and conservation checks. The resulting cumulative secretion deficits were only 3.42% at 5% AE4 and 3.86% in the null. Intracellular sodium rose transiently after AE4 loss but ended below wild type. Intracellular chloride remained lower, but this was not sufficient to sustain a large fluid deficit. NKCC1 increased its positive chloride loading by 20.82% and 23.16% in the two mutant cases and replaced most of the missing AE4 input.

The time course is shown by the dashed curves in figure 3. The absolute cumulative deficit continues to grow, but the percentage difference settles near 4%. Thus, after the acid–base and transporter reconstruction, cation routing alone no longer explains the organ-level phenotype that the 2018 model reproduced.

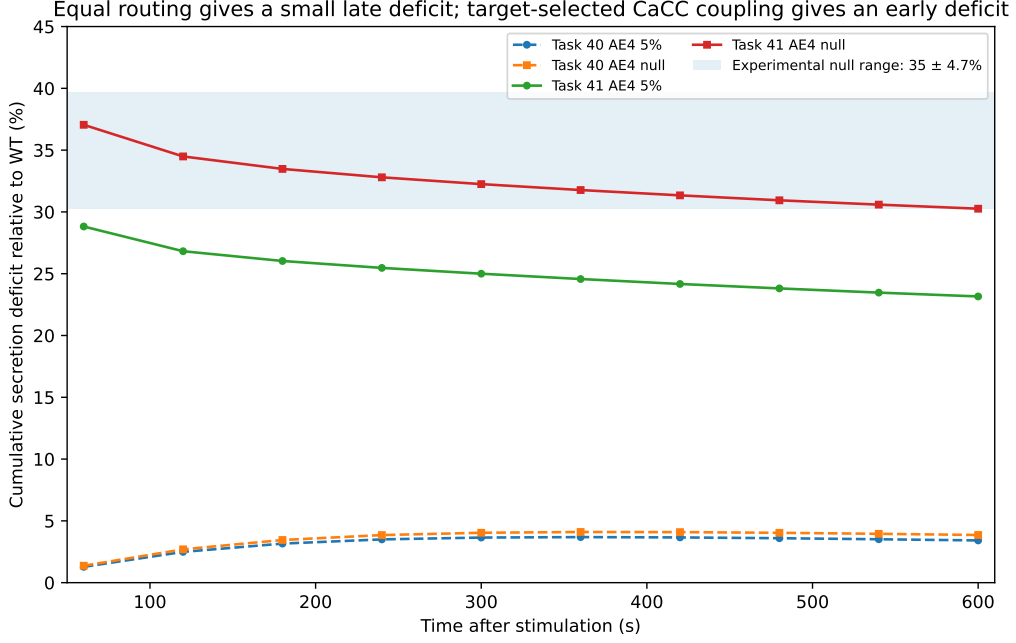


Figure 3: Cumulative secretion deficit relative to the corresponding wild type. Equal AE4 cation routing in Task 40 gives only a small sustained deficit. The target-selected CaCC construction in Task 41 reaches the experimental magnitude, but its percentage deficit is already largest early and then decreases. The shaded band is the reported AE4-null reduction of 35 ± 4.7 percentage points.

3.3 The steady salt balance removes the explicit equal-routing AE4 term

The small phenotype can be understood without inspecting every trajectory. Let $P = P_a + P_b$ denote the total sodium–potassium pump cycle rate. Under equal AE4 routing, combining the sodium, alkalinity and chloride equations in equation (4) eliminates NKCC1, AE2, NBC and the explicit AE4 cycle. The exact transient identity is

$$J_{Cl,a} = 6P - J_H + 2\dot{n}_{Na,i} - \dot{n}_{A,i} - \dot{n}_{Cl,i}. \quad (14)$$

At stationary state, equation (14) becomes

$$J_{Cl,a} = 6P - J_H. \quad (15)$$

AE4 still affects the state and therefore the pump, NHE1 and other transporters, but its cation split does not appear as an independent control term in the sustained apical chloride balance. This explains why changing the sodium–potassium routing can alter intracellular concentrations while leaving the secretion phenotype small.

The full cell-plus-lumen chloride identity adds another constraint,

$$q_{out}[Cl^-]_l = 2J_N + J_4 + J_2 - J_{Cl}^p - \frac{d}{dt}(n_{Cl,i} + n_{Cl,l}). \quad (16)$$

A missing AE4 chloride flux is therefore not equal to a fluid deficit. NKCC1 compensation, paracellular return, luminal composition and release from stored cellular chloride all contribute. Attempts to force the phenotype by imposing a finite NKCC1 ceiling confirmed this point. A strong ceiling generated a substantial algebraic loss but depleted intracellular chloride and pushed pH outside the inherited range; a physiology-constrained ceiling produced only an 8.6% sustained loss. During a 600-s simulation, stored chloride maintained secretion even as the null cell approached the chloride floor.

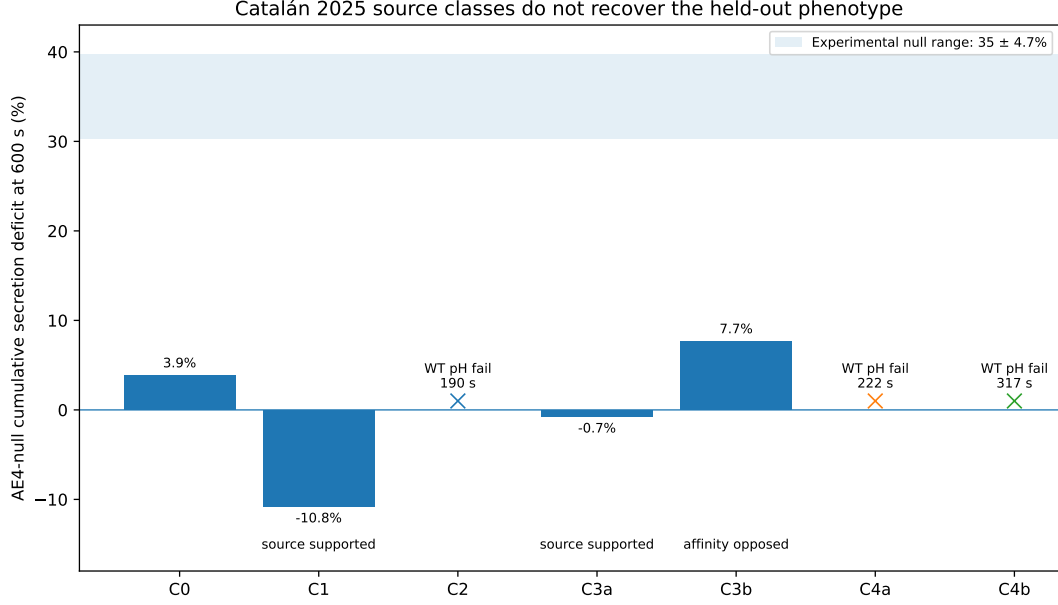


Figure 4: AE4-null secretion outcomes for the seven Catalán 2025 source classes. Negative deficits denote increased secretion after AE4 loss. C2, C4a and C4b do not have a valid 600-s wild-type reference because pH crossed the lower physiological gate during stimulation. None of the source-supported classes with a valid wild type approaches the held-out phenotype.

3.4 Catalán 2025 stoichiometric classes do not recover the phenotype

The seven classes in table 1 were evaluated as a finite, source-constrained panel. All source and charge tests passed. At the Task 40 reference state, the proposed forward direction was thermodynamically supported for C1, C2, C3a and C4a and opposed for C0, C3b and C4b. This mismatch is important because the first discrimination stage deliberately kept the inherited scalar J_4 even where its direction did not agree with the class-specific affinity.

All seven classes admitted a resting wild-type state. We then attempted wild type, 5% AE4 and null production trajectories from each frozen rest. Eighteen of the 21 trajectories passed. The wild-type trajectories for the three carbonate classes C2, C4a and C4b crossed the inherited lower pH gate at approximately 190, 222 and 317 s, respectively. These failures were retained without rescue.

None of the remaining source-supported classes recovered the held-out phenotype. C1, the proposed inward sodium–chloride and outward potassium–bicarbonate cycle, predicted a 10.76% *increase* in secretion after complete AE4 loss, accompanied by a 95.48% increase in NKCC1 chloride loading. C3a predicted essentially no phenotype, with a 0.72% secretion increase in the null. The largest valid secretion loss in the entire panel was 7.67% for C3b, but the forward direction of this class was opposed at the reference state. The complete comparison is shown in figure 4.

The result does not contradict the molecular experiments. Catalán et al. demonstrated cation dependence and identified residues that distinguish sodium-supported from potassium-supported transport; the whole-cycle stoichiometries remain proposed mechanisms [Catalán et al., 2025]. Our calculation shows only that changing transported source coefficients, while preserving the inherited Task 40 scalar kinetic law and the rest of the whole-cell network, is insufficient. The pH failures also show why the acid–base part of the model cannot be treated as an afterthought: a mechanism may be electroneutral, admit a resting solution and still fail during secretion because it removes carbon and alkalinity in the wrong proportion.

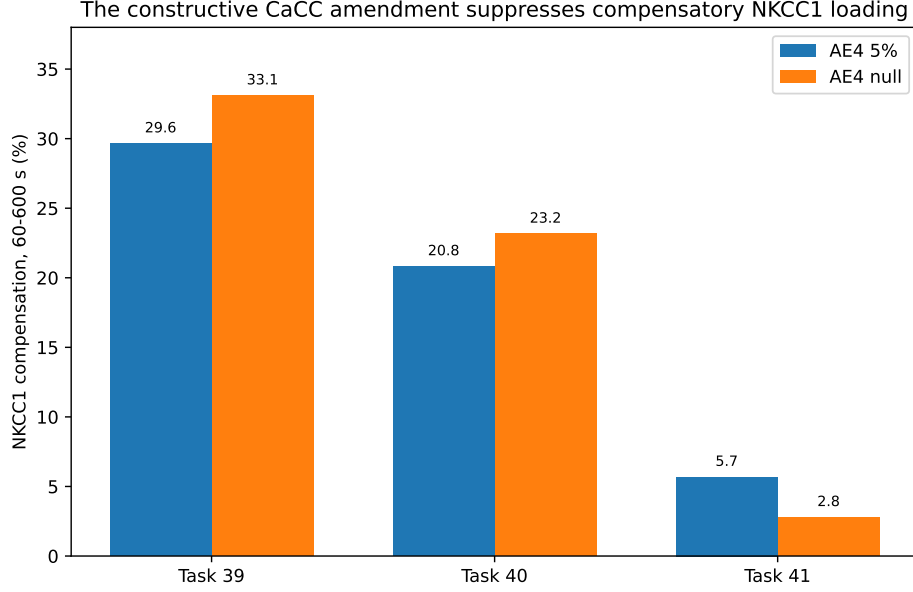


Figure 5: Integrated NKCC1 chloride-loading compensation after AE4 reduction. Source-fixed NKCC1 kinetics and equal routing permit strong compensation. The target-selected CaCC construction retains more intracellular chloride and substantially reduces the compensatory NKCC1 response.

3.5 A target-selected CaCC coupling reconstructs the magnitude but not the temporal signature

Having found that transporter routing and stoichiometry were insufficient, we asked a different question: what additional coupling can produce a substantial secretion loss without depleting intracellular chloride? This was an inverse design problem, not a held-out validation. Several static capacity changes were rejected. Limiting NKCC1 alone either gave too small a loss or produced unphysiological chloride and pH. Pump-affinity changes improved alkalinity but did not prevent chloride depletion. A static channel-conductance balance did not yield an admissible algebraic solution.

The successful construction makes the stimulated fraction of CaCC conductance depend on AE4 expression,

$$G_{Cl,a} = G_{Cl,a}^{\text{parent}} [(1 - u) + u\{b + (1 - b)e_4\}], \quad b = 0.1051187284, \quad (17)$$

where u is the existing normalised calcium activation and e_4 is relative AE4 expression. Wild type is exactly unchanged. At full stimulation, the conductance is 14.99% of the parent value at 5% AE4 and 10.51% in the null. The coupling is therefore strong: approximately 89.5% of fully stimulated recruitment is assumed to depend on AE4.

The construction gives cumulative secretion losses of 23.16% at 5% AE4 and 30.26% in the null while passing the declared 600-s gates. It also reduces NKCC1 compensation to 5.68% and 2.79%, compared with 20.82% and 23.16% under equal routing and 29.65% and 33.10% under the preceding source-fixed NKCC1 model. This progression is shown in figure 5.

However, the construction does not reproduce all independent observations. Intracellular sodium remains below wild type rather than increasing. Mutant chloride and cell volume finish above their wild-type values. The null pH is still rising at 600 s, so long-time validity has not been established. Most importantly, the percentage secretion deficit is largest within the first seconds of stimulation and then declines towards 30.26%, whereas the experimental phenotype

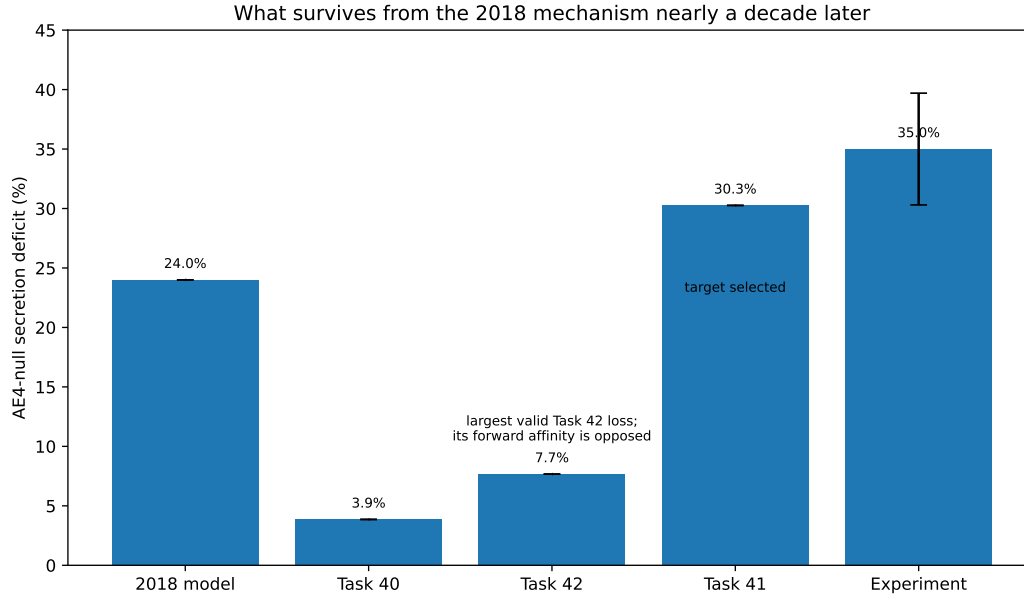


Figure 6: Summary of the AE4-null secretion phenotype across the original and reconstructed models. Task 42 shows the largest valid deficit among the tested source classes, but that class has an opposed forward affinity. Task 41 is an explicitly target-selected construction and must not be interpreted as an independent recovery of the phenotype.

develops during sustained secretion. The model therefore demonstrates that reduced apical chloride recruitment is mathematically capable of producing the required magnitude, but it does not establish AE4-dependent CaCC regulation as the biological mechanism.

3.6 What survives from the 2018 result

The sequence of models is summarised in figure 6. The 2018 model produced an approximately 24% AE4-null deficit. The conservation-explicit equal-routing model produces only 3.86%. The largest valid Task 42 deficit is 7.67%, and belongs to a class whose forward affinity is opposed at the reference state. The target-selected CaCC construction reaches 30.26%, close to the experimental central estimate of 35%, but by design rather than prediction.

The biological conclusion that AE4 is special survives. The original explanation that its cation routing alone creates the secretory phenotype does not. In the reconstructed system, acid-base supply determines whether AE4 can carry a substantial wild-type chloride load, and network compensation determines whether its removal matters for secretion.

4 Discussion

Nearly a decade ago, the first AE4 secretion model did two useful things. It showed that the apparently paradoxical AE4 and AE2 knockout experiments could be represented in one quantitative cell model, and it identified cation coupling as a plausible reason why AE4 could not be replaced by AE2 [Vera-Sigüenza et al., 2018]. The present reconstruction does not erase those conclusions. Instead, it separates the parts that survive from the parts that depended on the original model architecture.

4.1 The right transporter, but an incomplete network mechanism

The strongest conclusion of the 2018 work survives: AE4 is not simply a redundant copy of AE2. Experiments published since then have strengthened that point. AE4 requires a permeant monovalent cation, sodium and potassium do not interact with the transporter identically, and beta-adrenergic signalling regulates its activity through cAMP and PKA [Peña Münzenmayer et al., 2016, 2021, Catalán et al., 2025]. The original model therefore identified the correct biological actor and emphasised a property, cation dependence, that subsequent molecular work has confirmed.

What does not survive is the simple causal chain in which the AE4 cation split itself determines the gland phenotype. Once the surrounding acid–base system, NKCC1 law and NHE1 law are reconstructed, equal sodium and potassium routing gives only a 3–4% secretion deficit. The exact balance in equation (15) explains why. Under equal routing, the explicit AE4 term disappears from the sustained apical chloride balance. AE4 still changes intracellular states and thereby affects the pump, NHE1 and other fluxes, but the sodium–potassium split is not an independent control knob for secretion.

This distinction matters beyond the present model. A transporter can be biologically essential without appearing explicitly in a reduced steady balance. Its influence may be mediated through the state that determines other transporters, through regulatory coupling or through the availability of a conserved substrate. Conversely, reproducing a knockout phenotype does not prove that the omitted terms have been represented correctly. The 2018 agreement with flow was therefore informative but not mechanistically unique.

4.2 The transporter we removed turns out to be structurally necessary

The most consequential revision is the inward bicarbonate pathway. The 2018 model stated explicitly that no other proton or bicarbonate transport mechanism was included. That simplification was sufficient for the original calibration, but it becomes restrictive once AE4 is required to carry a substantial stimulated chloride load. Every original AE4 cycle exports two bicarbonates. In conserved coordinates, that is a loss of two carbon units and two alkalinity equivalents. NHE1 replaces alkalinity but not carbon, and the flux compatible with the measured sodium and pH state is not large enough to support the desired AE4 load.

The NBC-like pathway supplies the missing degree of freedom. It imports one sodium and two bicarbonates, enters the membrane-current closure because it is electrogenic, and is recruited only during stimulation. The reconstructed wild type can then assign approximately 20% of positive chloride loading to AE4 while keeping pH within the physiological interval. In plain terms, the pathway matters not because it secretes chloride itself, but because it supplies the bicarbonate that AE4 spends.

We do not know why NBC was removed during development of the original model, and it would be unhelpful to manufacture a retrospective explanation. The original paper used a minimal acid–base architecture. The present result shows where that minimality stops being harmless. This is a useful example of why model improvement is not always achieved by adding more detail everywhere. One particular omitted process becomes necessary because it closes a conserved balance that the original variables obscured.

The NBC implementation should nevertheless be interpreted carefully. It is an electrogenic $1\text{Na}^+ : 2\text{HCO}_3^-$ structural surrogate and not a claim that one specific NBC isoform, abundance or kinetic law has been identified in mouse submandibular acinar cells. Its capacity was derived from a wild-type feasibility calculation, not measured directly. Experiments that resolve sodium–bicarbonate transport during combined cholinergic and beta-adrenergic stimulation would therefore be especially valuable.

4.3 pH is a mechanism discriminator, not merely a calibration target

The Task 42 panel provides the clearest demonstration of the role of pH. All seven AE4 source classes were electroneutral and admitted an acceptable resting state. Three nevertheless failed during stimulation because pH crossed the lower physiological bound. The failures were confined to carbonate-export classes, for which one transported carbon unit removes two alkalinity equivalents. Resting admissibility and electroneutrality were therefore insufficient to establish physiological viability.

This result is directly relevant to the mechanism proposals of Catalán et al. [2025]. Their experiments establish distinct sodium- and potassium-dependent behaviour and identify residues involved in cation and bicarbonate transport. They do not resolve the complete stoichiometric cycle in the salivary acinar cell. Our whole-cell tests are compatible with those experiments: none disputes the cation dependence or mutant hierarchy. Instead, they show that some proposed source classes have acid–base consequences that become apparent only when the transporter is embedded in a secreting cell.

The result also illustrates why the Catalán classes cannot simply replace the Task 40 source law and be judged by a single flow number. C1 is thermodynamically supported at the reference state, but AE4 deletion increases secretion because NKCC1 nearly doubles its compensatory chloride loading. C3a is also supported but produces almost no secretion phenotype. C3b gives the largest valid deficit, but its proposed forward direction is opposed at the reference state. The molecular mechanism and the whole-cell network must therefore be considered together.

4.4 Compensation, stored chloride and sustained secretion

The reconstructed model repeatedly compensates for AE4 loss through NKCC1. This compensation is not a numerical curiosity. NKCC1 responds to the intracellular sodium, potassium and chloride state, and reduced AE4 loading changes all three. A static comparison of transporter capacities therefore does not determine the integrated chloride supply.

This prediction needs an important experimental qualification. In the 2015 study, NKCC1-dependent chloride uptake was isolated in bicarbonate-free conditions with the anion-exchange and apical chloride pathways suppressed, and no difference in NKCC1 functional activity was detected between control and AE4- or AE2-deficient acinar cells [Peña Münzenmayer et al., 2015]. The same study found no genotype-dependent change in the stimulation-induced NHE-dependent alkalisation rate or in resting intracellular pH. The approximately 23% increase in integrated NKCC1 chloride loading in the Task 40 AE4-null simulation is a different quantity: it is the modelled integral over 600 s of combined stimulation in a network with bicarbonate transport and changing intracellular state. The two observations are therefore not formally contradictory, but they are in clear experimental tension. We treat strong NKCC1 compensation as a model mechanism that requires direct testing, not as an established response to AE4 loss.

Stored intracellular chloride adds a second layer. Restricting NKCC1 can reduce sustained loading, but the cell may continue secreting for several minutes by releasing its existing chloride pool. A model may therefore reach the desired cumulative secretion deficit only by approaching an unphysiological chloride state. This is precisely what occurred in the finite-capacity tests. The phenotype cannot be assessed from one steady algebraic balance while ignoring the finite intracellular reservoir.

The experimental temporal pattern is particularly informative. AE4-deficient glands initially secrete at a rate close to wild type and diverge during sustained stimulation [Peña Münzenmayer et al., 2015]. C0 and C3b reproduce the direction of this timing but not its magnitude. The Task 41 CaCC construction reproduces the magnitude but not the timing,

because its deficit is already large early. A successful mechanism must satisfy both constraints. It must preserve the initial use of stored chloride while producing a sustained loading or regulatory shortfall that compensation does not erase.

4.5 What the constructive CaCC model does and does not show

The AE4-dependent CaCC recruitment law in equation (17) was selected using the requested 20–35% secretion range. It is therefore not a prediction and should not be compared with the Task 42 classes as though all models were tested on equal terms. Its purpose is different. It proves that a substantial secretion loss can be generated without forcing intracellular chloride below the physiological range, provided that AE4 loss also reduces sustained apical chloride exit strongly enough to weaken NKCC1 compensation.

The required dependency is large. The null retains only about 10.5% of the parent stimulated CaCC conductance. Such a strong relationship should be experimentally visible. Time-resolved measurements of TMEM16A/CaCC current under combined cholinergic and beta-adrenergic stimulation therefore provide a direct test. If sustained CaCC recruitment is close to wild type in AE4-deficient cells, the construction is falsified. If initial current is preserved but sustained recruitment declines sharply, the missing coupling becomes biologically plausible.

Other failures remain. The model does not reproduce higher intracellular sodium after AE4 loss, mutant cells end with larger volume, and long-time pH has not been validated beyond 600 s. The constructive model should therefore be viewed as a local statement about network architecture: some reduction in sustained chloride exit can reconcile secretion magnitude with chloride retention, but the particular AE4–CaCC law is not established.

4.6 A ten-year reassessment rather than a replacement

The present work is best understood as a reassessment of the original explanation, not a repudiation of the original paper. The 2018 model was built with the information and computational architecture available at the time. It correctly highlighted AE4 and its cation dependence. The new model adds constraints that were previously absent: conserved carbon and alkalinity, a finite buffer, electrogenic bicarbonate entry, source-fixed NHE1 and NKCC1 laws, beta/cAMP/PKA regulation and molecularly motivated AE4 source classes.

With those constraints, the original phenotype is harder to reproduce. That is scientific progress rather than failure. A less constrained model can agree with one observation for several reasons. A more constrained model reveals which reasons remain possible. Here, the biological importance of AE4 remains, but the original cation-routing explanation is insufficient. The missing mechanism lies in the interaction between acid–base supply, compensatory chloride loading and sustained regulation.

4.7 Scope and limitations

Several limits on the interpretation should be explicit. First, the model represents one well-stirred acinar cell coupled to a finite lumen and a fixed extracellular bath. It describes primary acinar fluid secretion, not ductal modification of saliva, yet the knockout phenotype used for comparison is organ-level submandibular gland flow. Agreement or disagreement with that benchmark therefore cannot by itself identify every cellular contribution to the intact gland.

Second, calcium is prescribed rather than generated by a coupled signalling model, and beta-adrenergic control is represented by a deliberately small effective regulatory system. Ionic state, cell volume and pH do not feed back onto the imposed calcium waveform. The reconstructed NHE1 law is imported from an established kinetic model rather than fitted

to salivary acinar measurements, while AE2 is represented by a reversible thermodynamic surrogate. The NBC-like pathway is likewise structural: the present work establishes the need for an inward bicarbonate-equivalent source within this model architecture, not the molecular identity, abundance or detailed kinetics of a particular salivary NBC isoform.

Third, Task 42 changes the AE4 source class while deliberately retaining the inherited scalar AE4 kinetics and the rest of the whole-cell network. Its negative result therefore concerns those seven stoichiometric source substitutions under fixed kinetics. It does not exclude a mechanism in which the 2025 molecular determinants also alter affinities, rate constants, regulation or interactions with other transporters. Conversely, Task 41 is target-selected and demonstrates sufficiency of one additional coupling, not evidence that AE4 controls CaCC recruitment by the fitted law.

Finally, the physiological gates were evaluated over the declared 600 s window. Passing them establishes finite-horizon consistency with those criteria only. It does not establish long-time stability, chronic compensation or complete physiological validity. These distinctions are particularly important here because the experimental phenotype develops over minutes and because the central unresolved question is itself a question of sustained regulation.

4.8 Experimental predictions and next steps

The model identifies several measurements that would discriminate the remaining possibilities.

First, stimulated sodium–bicarbonate transport should be quantified in acinar cells under the same combined stimulation used for secretion. The model predicts that a substantial inward bicarbonate-equivalent flux is required if AE4 contributes materially to chloride loading while pH remains near neutral.

Second, NKCC1 activity should be measured through time rather than only as an end-point or isolated uptake rate. The competing models predict markedly different integrated compensation after AE4 loss.

Third, intracellular sodium, chloride, pH and volume should be recorded together. These variables discriminate models that produce similar total secretion by very different internal routes.

Fourth, apical chloride conductance should be resolved into its early and sustained components in AE4-deficient cells. This is the direct test of the constructive CaCC coupling.

Finally, the Catalán mechanism classes require class-specific kinetic laws. Task 42 deliberately preserved the inherited scalar J_4 to isolate source stoichiometry. A molecular model that couples the newly identified residues to rate constants, affinities and regulation could change the whole-cell conclusions. The present negative result applies to stoichiometric source substitution under the inherited kinetics, not to every mechanistic realisation compatible with the 2025 experiments.

5 Conclusion

The original AE4 model identified the correct biological problem: AE4 is not interchangeable with AE2, and its cation dependence matters. Nearly a decade later, however, a conservation-explicit reconstruction shows that cation stoichiometry alone does not explain the secretion phenotype. An inward bicarbonate pathway is required for substantial wild-type AE4 chloride loading without loss of pH control. Once that pathway and modern NKCC1 and NHE1 laws are included, the network compensates strongly for AE4 loss. None of the source classes proposed from the 2025 molecular experiments recovers the approximately 35% deficit under the inherited whole-cell kinetics, and several are rejected by stimulated pH rather than by secretion alone.

A target-selected reduction in stimulated CaCC recruitment can reconstruct the phenotype magnitude, but not its complete ionic and temporal signature. The decisive unresolved question is therefore no longer simply what AE4 transports. It is how AE4 transport, acid–base supply and sustained apical chloride exit are regulated together during secretion.

Data and code availability

All model source, frozen prediction checkpoints, compact trajectories, verification records and figure-building scripts are contained in the accompanying public repository. The manuscript figures are regenerated from `manuscript/data/figure_source_values.csv` by `manuscript/scripts/build_figures.py`. The Task 42 mechanism-class predictions were committed before the held-out phenotype comparison.

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